

In [1]: `!pip install ydata-profiling`

```
import pandas as pd
```

```
from ydata_profiling import ProfileReport
```

```
# Now you can use the profiling tool
```

```
# Example usage:
```

```
# df = pd.DataFrame({'col1': [1, 2, 3], 'col2': [4, 5, 6]})
```

```
# profile = ProfileReport(df, title="Pandas Profiling Report")
```

```
# profile.to_file("output.html")
```

Collecting ydata-profiling

Using cached ydata_profiling-4.18.1-py2.py3-none-any.whl.metadata (22 kB)

Requirement already satisfied: scipy<1.17,>=1.8 in c:\users\lenovo\anaconda3\lib\site-packages (from ydata-profiling) (1.15.3)

Requirement already satisfied: pandas!=1.4.0,<3.0,>1.5 in c:\users\lenovo\anaconda3\lib\site-packages (from ydata-profiling) (2.2.3)

Requirement already satisfied: matplotlib<=3.10,>=3.5 in c:\users\lenovo\anaconda3\lib\site-packages (from ydata-profiling) (3.10.0)

Requirement already satisfied: pydantic<3,>=2 in c:\users\lenovo\anaconda3\lib\site-packages (from ydata-profiling) (2.10.3)

Collecting PyYAML<6.1,>=6.0.3 (from ydata-profiling)

Using cached pyyaml-6.0.3-cp313-cp313-win_amd64.whl.metadata (2.4 kB)

Requirement already satisfied: jinja2<3.2,>=3.1.6 in c:\users\lenovo\anaconda3\lib\site-packages (from ydata-profiling) (3.1.6)

Collecting visions<0.8.2,>=0.7.5 (from visions[type_image_path]<0.8.2,>=0.7.5->ydata-profiling)

Using cached visions-0.8.1-py3-none-any.whl.metadata (11 kB)

Requirement already satisfied: numpy<2.4,>=1.22 in c:\users\lenovo\anaconda3\lib\site-packages (from ydata-profiling) (2.1.3)

Collecting minify-html>=0.15.0 (from ydata-profiling)

Using cached minify_html-0.18.1-cp313-cp313-win_amd64.whl.metadata (18 kB)

Collecting filetype>=1.0.0 (from ydata-profiling)

Using cached filetype-1.2.0-py2.py3-none-any.whl.metadata (6.5 kB)

Collecting phik<0.13,>=0.12.5 (from ydata-profiling)

Using cached phik-0.12.5-cp313-cp313-win_amd64.whl.metadata (5.6 kB)

Requirement already satisfied: requests<3,>=2.32.0 in c:\users\lenovo\anaconda3\lib\site-packages (from ydata-profiling) (2.32.3)

Requirement already satisfied: tqdm<5,>=4.66.3 in c:\users\lenovo\anaconda3\lib\site-packages (from ydata-profiling) (4.67.1)

Requirement already satisfied: seaborn<0.14,>=0.10.1 in c:\users\lenovo\anaconda3\lib\site-packages (from ydata-profiling) (0.13.2)

Collecting multimethod<2,>=1.4 (from ydata-profiling)

Using cached multimethod-1.12-py3-none-any.whl.metadata (9.6 kB)

Requirement already satisfied: statsmodels<1,>=0.13.2 in c:\users\lenovo\anaconda3\lib\site-packages (from ydata-profiling) (0.14.4)

Collecting typeguard<5,>=4 (from ydata-profiling)

Using cached typeguard-4.4.4-py3-none-any.whl.metadata (3.3 kB)

Collecting imagehash==4.3.2 (from ydata-profiling)

Using cached ImageHash-4.3.2-py2.py3-none-any.whl.metadata (8.4 kB)

Collecting wordcloud>=1.9.4 (from ydata-profiling)

Using cached wordcloud-1.9.6-cp313-cp313-win_amd64.whl.metadata (3.5 kB)

Collecting dacite<2,>=1.9 (from ydata-profiling)

Using cached dacite-1.9.2-py3-none-any.whl.metadata (17 kB)

Requirement already satisfied: numba<0.63,>=0.60 in c:\users\lenovo\anaconda3\lib\site-packages (from ydata-profiling) (0.61.0)

Requirement already satisfied: PyWavelets in c:\users\lenovo\anaconda3\lib\site-packages (from imagehash==4.3.2->ydata-profiling) (1.8.0)

Requirement already satisfied: pillow in c:\users\lenovo\anaconda3\lib\site-packages (from imagehash==4.3.2->ydata-profiling) (11.1.0)

Requirement already satisfied: MarkupSafe>=2.0 in c:\users\lenovo\anaconda3\lib\site-packages (from jinja2<3.2,>=3.1.6->ydata-profiling) (3.0.2)

Requirement already satisfied: contourpy>=1.0.1 in c:\users\lenovo\anaconda3\lib\site-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (1.3.1)

Requirement already satisfied: cycler>=0.10 in c:\users\lenovo\anaconda3\lib\site-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (0.11.0)

Requirement already satisfied: fonttools>=4.22.0 in c:\users\lenovo\anaconda3\lib\site-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (4.55.3)

Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\lenovo\anaconda3\lib\site-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (1.4.8)

Requirement already satisfied: packaging>=20.0 in c:\users\lenovo\anaconda3\lib\site-packages (from ydata-profiling) (24.0)

```

ite-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (24.2)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\lenovo\anaconda3\lib
\site-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (3.2.0)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\lenovo\anaconda3
\lib\site-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (2.9.0.post0)
Requirement already satisfied: llvmlite<0.45,>=0.44.0dev0 in c:\users\lenovo\anac
onda3\lib\site-packages (from numba<0.63,>=0.60->ydata-profiling) (0.44.0)
Requirement already satisfied: pytz>=2020.1 in c:\users\lenovo\anaconda3\lib\site
-packages (from pandas!=1.4.0,<3.0,>1.5->ydata-profiling) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\lenovo\anaconda3\lib\si
te-packages (from pandas!=1.4.0,<3.0,>1.5->ydata-profiling) (2025.2)
Requirement already satisfied: joblib>=0.14.1 in c:\users\lenovo\anaconda3\lib\si
te-packages (from phik<0.13,>=0.12.5->ydata-profiling) (1.4.2)
Requirement already satisfied: annotated-types>=0.6.0 in c:\users\lenovo\anaconda
3\lib\site-packages (from pydantic<3,>=2->ydata-profiling) (0.6.0)
Requirement already satisfied: pydantic-core==2.27.1 in c:\users\lenovo\anaconda3
\lib\site-packages (from pydantic<3,>=2->ydata-profiling) (2.27.1)
Requirement already satisfied: typing-extensions>=4.12.2 in c:\users\lenovo\anaco
nda3\lib\site-packages (from pydantic<3,>=2->ydata-profiling) (4.12.2)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\lenovo\anacon
da3\lib\site-packages (from requests<3,>=2.32.0->ydata-profiling) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in c:\users\lenovo\anaconda3\lib\site
-packages (from requests<3,>=2.32.0->ydata-profiling) (3.7)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\lenovo\anaconda3\li
b\site-packages (from requests<3,>=2.32.0->ydata-profiling) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\lenovo\anaconda3\li
b\site-packages (from requests<3,>=2.32.0->ydata-profiling) (2025.8.3)
Requirement already satisfied: patsy>=0.5.6 in c:\users\lenovo\anaconda3\lib\site
-packages (from statsmodels<1,>=0.13.2->ydata-profiling) (1.0.1)
Requirement already satisfied: colorama in c:\users\lenovo\anaconda3\lib\site-pac
kages (from tqdm<5,>=4.66.3->ydata-profiling) (0.4.6)
Collecting typing-extensions>=4.12.2 (from pydantic<3,>=2->ydata-profiling)
  Using cached typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
Requirement already satisfied: attrs>=19.3.0 in c:\users\lenovo\anaconda3\lib\sit
e-packages (from visions<0.8.2,>=0.7.5->visions[type_image_path]<0.8.2,>=0.7.5->y
data-profiling) (24.3.0)
Requirement already satisfied: networkx>=2.4 in c:\users\lenovo\anaconda3\lib\sit
e-packages (from visions<0.8.2,>=0.7.5->visions[type_image_path]<0.8.2,>=0.7.5->y
data-profiling) (3.4.2)
Collecting puremagic (from visions<0.8.2,>=0.7.5->visions[type_image_path]<0.8.2,
>=0.7.5->ydata-profiling)
  Using cached puremagic-1.30-py3-none-any.whl.metadata (5.8 kB)
Requirement already satisfied: six>=1.5 in c:\users\lenovo\anaconda3\lib\site-pac
kages (from python-dateutil>=2.7->matplotlib<=3.10,>=3.5->ydata-profiling) (1.17.
0)
Using cached ydata_profiling-4.18.1-py2.py3-none-any.whl (400 kB)
Using cached ImageHash-4.3.2-py2.py3-none-any.whl (296 kB)
Using cached dacite-1.9.2-py3-none-any.whl (16 kB)
Using cached multimethod-1.12-py3-none-any.whl (10 kB)
Using cached phik-0.12.5-cp313-cp313-win_amd64.whl (678 kB)
Using cached pyyaml-6.0.3-cp313-cp313-win_amd64.whl (154 kB)
Using cached typeguard-4.4.4-py3-none-any.whl (34 kB)
Using cached visions-0.8.1-py3-none-any.whl (105 kB)
Using cached filetype-1.2.0-py2.py3-none-any.whl (19 kB)
Downloading minify_html-0.18.1-cp313-cp313-win_amd64.whl (3.1 MB)
----- 0.0/3.1 MB ? eta -:--:--
----- 0.0/3.1 MB ? eta -:--:--
--- ----- 0.3/3.1 MB ? eta -:--:--
----- 0.5/3.1 MB 1.1 MB/s eta 0:00:03
----- 0.8/3.1 MB 1.0 MB/s eta 0:00:03

```

```

----- 1.0/3.1 MB 1.1 MB/s eta 0:00:02
----- 1.3/3.1 MB 1.2 MB/s eta 0:00:02
----- 1.6/3.1 MB 1.1 MB/s eta 0:00:02
----- 1.6/3.1 MB 1.1 MB/s eta 0:00:02
----- 1.8/3.1 MB 1.0 MB/s eta 0:00:02
----- 2.1/3.1 MB 1.0 MB/s eta 0:00:02
----- 2.1/3.1 MB 1.0 MB/s eta 0:00:02
----- 2.4/3.1 MB 982.4 kB/s eta 0:00:01
----- 2.6/3.1 MB 948.1 kB/s eta 0:00:01
----- 2.6/3.1 MB 948.1 kB/s eta 0:00:01
----- 2.9/3.1 MB 914.7 kB/s eta 0:00:01
----- 2.9/3.1 MB 914.7 kB/s eta 0:00:01
----- 3.1/3.1 MB 899.0 kB/s eta 0:00:00
Downloading typing_extensions-4.15.0-py3-none-any.whl (44 kB)
Downloading wordcloud-1.9.6-cp313-cp313-win_amd64.whl (306 kB)
Downloading puremagic-1.30-py3-none-any.whl (43 kB)
Installing collected packages: puremagic, minify-html, filetype, typing-extensions, PyYAML, multimethod, dacite, typeguard, imagehash, wordcloud, visions, phik, ydata-profiling

```


WARNING: Failed to remove contents in a temporary directory 'C:\Users\LENOVO\anaconda3\Lib\site-packages\~aml'.
You can safely remove it manually.

```
In [2]: %%time
import pandas as pd
import seaborn as sns
from matplotlib import pyplot as plt
import pandas_profiling as prf
```

CPU times: total: 31.2 ms

Wall time: 5.69 ms

<timed exec>:4: DeprecationWarning: `import pandas\_profiling` is going to be deprecated by April 1st. Please use `import ydata\_profiling` instead.

```
In [6]: netflix = pd.read_csv("D:\\manish\\Python\\ineuron\\Anand Jha Drive for INEuron\\netflix.csv")
netflix.info()
```

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 8807 entries, 0 to 8806

Data columns (total 13 columns):

#	Column	Non-Null Count	Dtype
0	Unnamed: 0	8807 non-null	int64
1	show_id	8807 non-null	object
2	type	8807 non-null	object
3	title	8807 non-null	object
4	director	6173 non-null	object
5	cast	7982 non-null	object
6	country	7976 non-null	object
7	date_added	8797 non-null	object
8	release_year	8807 non-null	int64
9	rating	8803 non-null	object
10	duration	8804 non-null	object
11	listed_in	8807 non-null	object
12	description	8807 non-null	object

dtypes: int64(2), object(11)

memory usage: 894.6+ KB

About Data

This dataset contains data collected from Netflix of different TV shows and movies from the year 2008 to 2021.

- type: Gives information about 2 different unique values one is TV Show and another is Movie
- title: Gives information about the title of Movie or TV Show
- director: Gives information about the director who directed the Movie or TV Show
- cast: Gives information about the cast who plays role in Movie or TV Show
- release_year: Gives information about the year when Movie or TV Show was released
- rating: Gives information about the Movie or TV Show are in which category (eg like the movies are only for students, or adults, etc)
- duration: Gives information about the duration of Movie or TV Show
- listed_in: Gives information about the genre of Movie or TV Show
- description: Gives information about the description of Movie or TV Show

```
In [7]: m=netflix.isnull().sum()
miss= (netflix.isnull().sum()/len(netflix))*100
miss_data=pd.concat([m,miss],axis=1,keys=['Total','%'])
print(miss_data)
```

	Total	%
Unnamed: 0	0	0.000000
show_id	0	0.000000
type	0	0.000000
title	0	0.000000
director	2634	29.908028
cast	825	9.367549
country	831	9.435676
date_added	10	0.113546
release_year	0	0.000000
rating	4	0.045418
duration	3	0.034064
listed_in	0	0.000000
description	0	0.000000

```
In [8]: netflix = netflix.dropna( how='any',subset=['cast', 'director'])
```

```
In [9]: netflix.fillna({ 'country': 'missing', 'rating': 'missing'}, inplace=True)
netflix.isnull().sum()
```

```
Out[9]: Unnamed: 0      0
show_id      0
type         0
title        0
director     0
cast         0
country      0
date_added   0
release_year  0
rating       0
duration     3
listed_in    0
description  0
dtype: int64
```

```
In [10]: netflix['date_added'] = pd.to_datetime(netflix['date_added'])
netflix.head()
```

C:\Users\LENOVO\AppData\Local\Temp\ipykernel_14916\3787484694.py:1: UserWarning: Could not infer format, so each element will be parsed individually, falling back to `dateutil`. To ensure parsing is consistent and as-expected, please specify a format.

```
netflix['date_added'] = pd.to_datetime(netflix['date_added'])
```

Out[10]:

Unnamed: 0	show_id	type	title	director	cast	country	date_adde
2	2	s3 TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	missing	2021-09-2
5	5	s6 TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	missing	2021-09-2
6	6	s7 Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	missing	2021-09-2
7	7	s8 Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...	United States, Ghana, Burkina Faso, United Kin...	2021-09-2
8	8	s9 TV Show	The Great British Baking Show	Andy Devonshire	Mel Giedroyc, Sue Perkins, Mary Berry, Paul Ho...	United Kingdom	2021-09-2

In [11]: `## Finding how many unique values are there in the dataset`
`netflix.nunique()`

Out[11]:

Unnamed: 0	5700
show_id	5700
type	2
title	5698
director	4152
cast	5512
country	604
date_added	1478
release_year	72
rating	18
duration	205
listed_in	346
description	5677
dtype:	int64

In [12]: `netflix.describe(include='all')`

Out[12]:

	Unnamed: 0	show_id	type	title	director	cast	country	date_added
count	5700.000000	5700	5700	5700	5700	5700	5700	5700
unique	NaN	5700	2	5698	4152	5512	604	1
top	NaN	s8807	Movie	22-Jul	Raúl Campos, Jan Suter	Vatsal Dubey, Julie Tejjwani, Rupa Bhimani, Jig...	United States	1
freq	NaN	1	5522	2	18	13	1849	1
mean	4551.407018	NaN	NaN	NaN	NaN	NaN	NaN	2019-05-02:41:56.210526
min	2.000000	NaN	NaN	NaN	NaN	NaN	NaN	2008-01-00:00:00
25%	2276.750000	NaN	NaN	NaN	NaN	NaN	NaN	2018-05-00:00:00
50%	4697.500000	NaN	NaN	NaN	NaN	NaN	NaN	2019-07-12:00:00
75%	6828.250000	NaN	NaN	NaN	NaN	NaN	NaN	2020-08-00:00:00
max	8806.000000	NaN	NaN	NaN	NaN	NaN	NaN	2021-09-00:00:00
std	2590.214753	NaN	NaN	NaN	NaN	NaN	NaN	1

EDA

- What different types of show or movie are uploaded on Netflix?
- Correlation between the features
- Most watched shows on the Netflix
- Distribution of Ratings
- Which has the highest rating Tv show or Movies
- Finding the best Month for releasing content
- Highest watched genres on Netflix
- Released movie over the years

```
In [13]: ##method-1
netflix.groupby('type')['title'].count().sort_values(ascending = False)
```

```
Out[13]: type
Movie      5522
TV Show     178
Name: title, dtype: int64
```

```
In [14]: ##method-2
netflix.type.value_counts().to_frame('values_count')
```

```
Out[14]:
```

values_count	
type	
Movie	5522
TV Show	178

```
In [16]: numeric_columns = netflix.select_dtypes(include=['float64', 'int64']).columns

correlation_matrix = netflix[numeric_columns].corr()

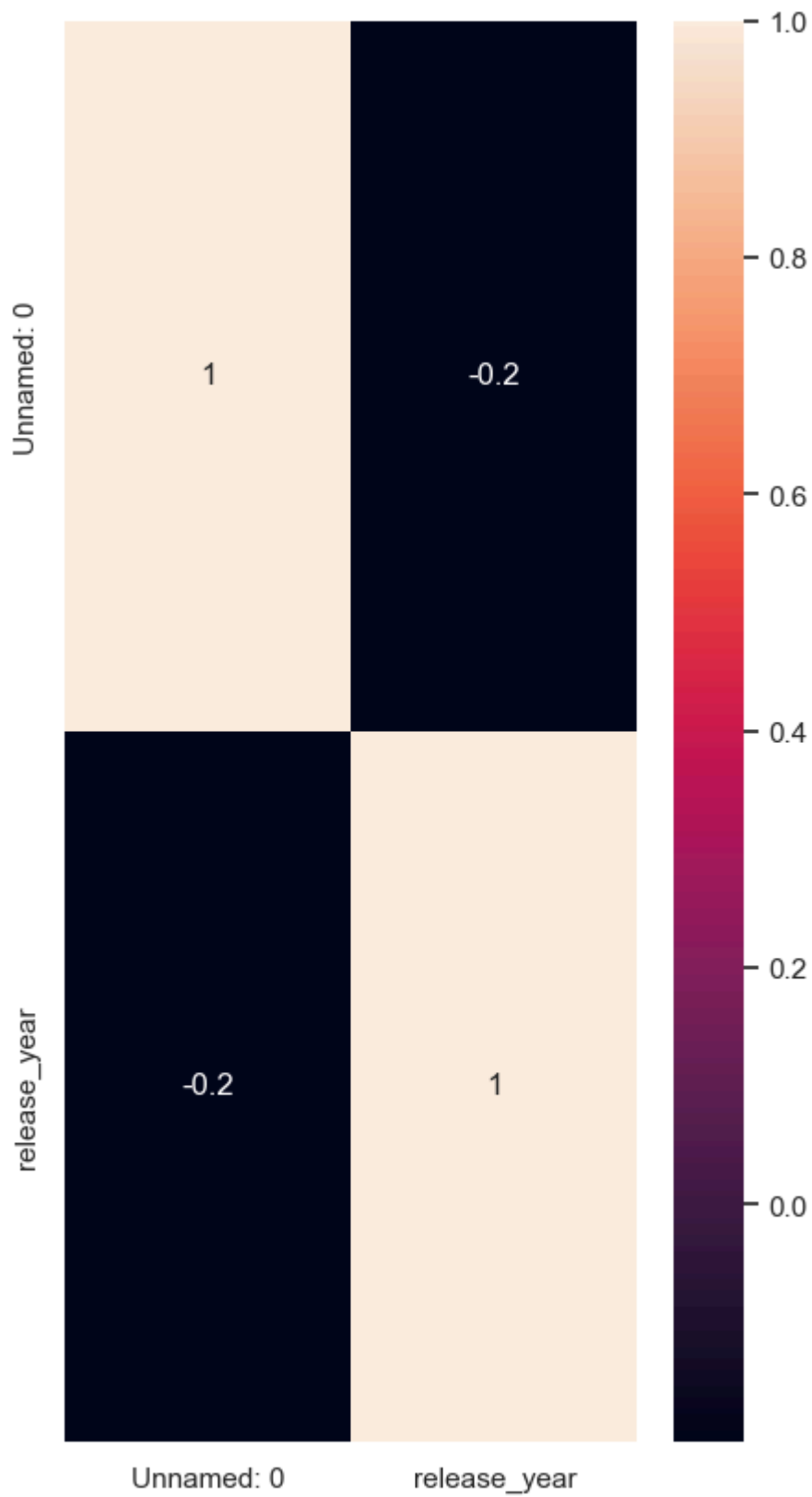
correlation_matrix
```

```
Out[16]:
```

	Unnamed: 0	release_year
Unnamed: 0	1.000000	-0.199716
release_year	-0.199716	1.000000

```
In [18]: sns.set()
plt.subplots(figsize=(5,10))
sns.heatmap(correlation_matrix,annot=True)
```

```
Out[18]: <Axes: >
```



```
In [19]: netflix_profile = prf.ProfileReport(netflix)
netflix_profile
```

Summarize dataset: 0% | 0/5 [00:00<?, ?it/s]



Overview

Overview

Alerts 3

Reproduction

Dataset statistics

Number of variables	13
Number of observations	5700
Missing cells	3
Missing cells (%)	< 0.1%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	623.4 KiB
Average record size in memory	112.0 B

Variable types

Numeric	2
Text	8
Categorical	2
DateTime	1



Out[19]:

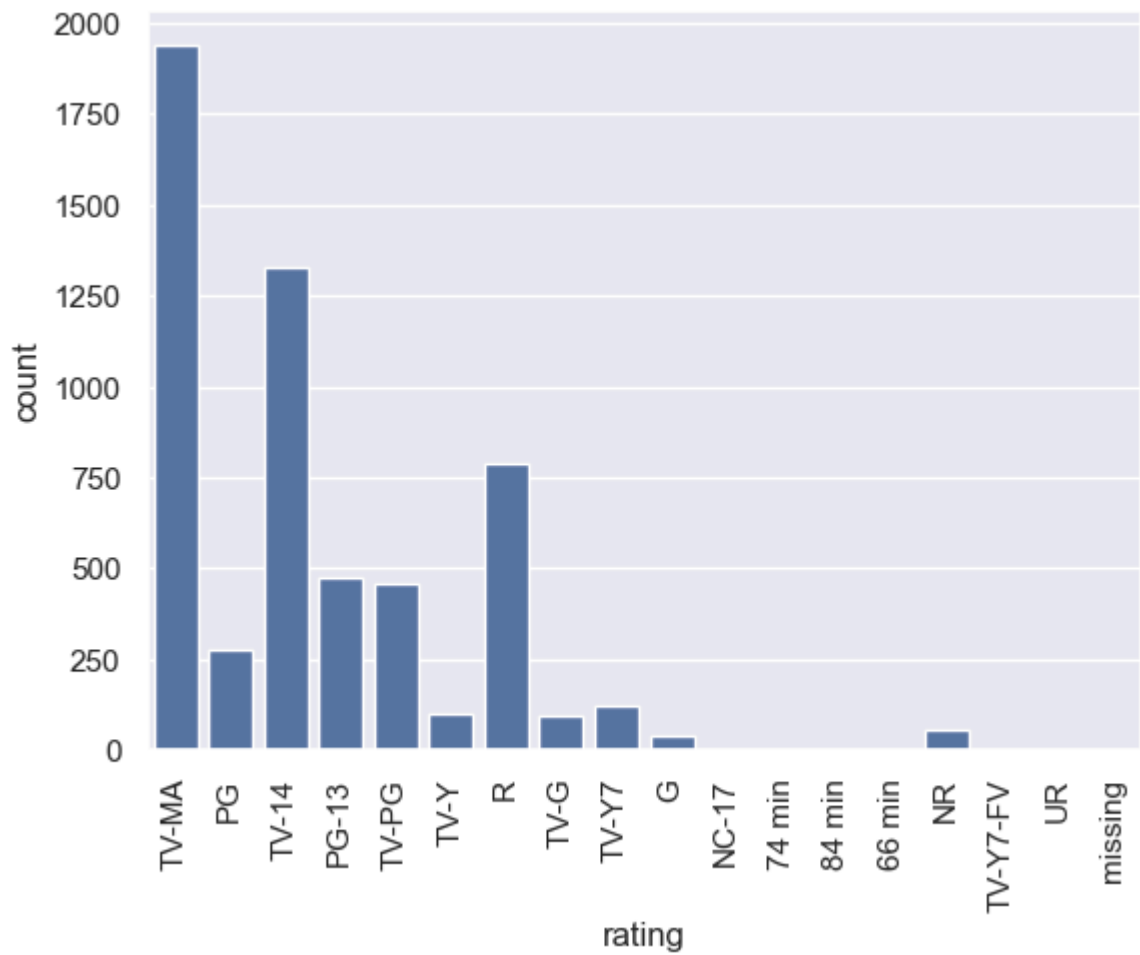
```
In [22]: netflix_profile.to_file(output_file="Movies_before_preprocessing.html")
Export report to file:  0%|          | 0/1 [00:00<?, ?it/s]

In [23]: netflix.groupby(["type", "release_year"])["title"].agg(pd.Series.mode)
```

```
Out[23]: type      release_year
Movie      1942          The Battle of Midway
           1944          Tunisian Victory
           1945          Know Your Enemy - Japan
           1946          Let There Be Light
           1947          Thunderbolt
           ...
TV Show    2017          [Abnormal Summit, Afronta! Facing It!, Ari Sha...
           2018          [A.I.C.O., Age of Rebellion, Bitter Daisies, B...
           2019          [7 (Seven), A Perfect Day For Arsenide, Abby H...
           2020          [A Queen Is Born, Almost Happy, Anitta: Made I...
           2021          [Bangkok Breaking, Clickbait, Dealer, Feel Goo...
Name: title, Length: 88, dtype: object
```

```
In [24]: sns.countplot(x=netflix['rating'],orient = 'v')
plt.xticks(rotation = 90)
```

```
Out[24]: ([0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17],
[Text(0, 0, 'TV-MA'),
 Text(1, 0, 'PG'),
 Text(2, 0, 'TV-14'),
 Text(3, 0, 'PG-13'),
 Text(4, 0, 'TV-PG'),
 Text(5, 0, 'TV-Y'),
 Text(6, 0, 'R'),
 Text(7, 0, 'TV-G'),
 Text(8, 0, 'TV-Y7'),
 Text(9, 0, 'G'),
 Text(10, 0, 'NC-17'),
 Text(11, 0, '74 min'),
 Text(12, 0, '84 min'),
 Text(13, 0, '66 min'),
 Text(14, 0, 'NR'),
 Text(15, 0, 'TV-Y7-FV'),
 Text(16, 0, 'UR'),
 Text(17, 0, 'missing')])
```



```
In [25]: netflix.groupby("type")["rating"].agg(pd.Series.mode)
```

```
Out[25]: type
Movie    TV-MA
TV Show  TV-MA
Name: rating, dtype: object
```

```
In [26]: netflix.head()
```

Out[26]:

	Unnamed: 0	show_id	type	title	director	cast	country	date_adde
2	2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	missing	2021-09-2
5	5	s6	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	missing	2021-09-2
6	6	s7	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	missing	2021-09-2
7	7	s8	Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...	United States, Ghana, Burkina Faso, United Kin...	2021-09-2
8	8	s9	TV Show	The Great British Baking Show	Andy Devonshire	Mel Giedroyc, Sue Perkins, Mary Berry, Paul Ho...	United Kingdom	2021-09-2

In [30]: `new_genre = netflix['listed_in'].str.split(",",n=2)`

In [31]: `new_genre`

```

Out[31]: 2      [Crime TV Shows, International TV Shows, TV ...
5          [TV Dramas, TV Horror, TV Mysteries]
6          [Children & Family Movies]
7      [Dramas, Independent Movies, International M...
8          [British TV Shows, Reality TV]
...
8801     [Dramas, International Movies, Thrillers]
8802     [Cult Movies, Dramas, Thrillers]
8804     [Comedies, Horror Movies]
8805     [Children & Family Movies, Comedies]
8806     [Dramas, International Movies, Music & Music...
Name: listed_in, Length: 5700, dtype: object

```

Adding new features


```
In [32]: netflix['Genre 1'] = new_genre.str.get(0)
netflix['Genre 2'] = new_genre.str.get(1)
netflix['Genre 3'] = new_genre.str.get(2)
```

```
In [33]: netflix.head()
```

```
Out[33]:
```

	Unnamed: 0	show_id	type	title	director	cast	country	date_adde
2	2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	missing	2021-09-2
5	5	s6	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	missing	2021-09-2
6	6	s7	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	missing	2021-09-2
7	7	s8	Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...	United States, Ghana, Burkina Faso, United Kin...	2021-09-2
8	8	s9	TV Show	The Great British Baking Show	Andy Devonshire	Mel Giedroyc, Sue Perkins, Mary Berry, Paul Ho...	United Kingdom	2021-09-2

```
In [34]: netflix['Genre 1'].describe(include='all')
```

```
Out[34]: count      5700
unique        33
top      Dramas
freq       1586
Name: Genre 1, dtype: object
```

```
In [35]: netflix['Genre 2'].describe(include='all')
```

```
Out[35]: count          4485
unique           35
top      International Movies
freq          1545
Name: Genre 2, dtype: object
```

```
In [36]: netflix['Genre 3'].describe(include='all')
```

```
Out[36]: count          2434
unique           29
top      International Movies
freq          881
Name: Genre 3, dtype: object
```

```
In [37]: netflix.drop('listed_in',axis = 1,inplace = True)
```

```
In [38]: netflix.head()
```

```
Out[38]:
```

	Unnamed: 0	show_id	type	title	director	cast	country	date_adde
2	2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	missing	2021-09-2
5	5	s6	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	missing	2021-09-2
6	6	s7	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	missing	2021-09-2
7	7	s8	Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...	United States, Ghana, Burkina Faso, United Kin...	2021-09-2
8	8	s9	TV Show	The Great British Baking Show	Andy Devonshire	Mel Giedroyc, Sue Perkins, Mary Berry, Paul Ho...	United Kingdom	2021-09-2

```
In [39]: netflix.drop_duplicates(subset = None,keep='first',inplace=False)
```

Out[39]:

Unnamed: 0	show_id	type	title	director	cast	country	date_
2	2	s3 TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	missing	2021-
5	5	s6 TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	missing	2021-
6	6	s7 Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	missing	2021-
7	7	s8 Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...	United States, Ghana, Burkina Faso, United Kin...	2021-
8	8	s9 TV Show	The Great British Baking Show	Andy Devonshire	Mel Giedroyc, Sue Perkins, Mary Berry, Paul Ho...	United Kingdom	2021-
...	...	...	...	...	...	...	...
8801	8801	s8802 Movie	Zinzana	Majid Al Ansari	Ali Suliman, Saleh Bakri, Yasa, Ali Al-Jabri, ...	United Arab Emirates, Jordan	2016-
8802	8802	s8803 Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-
8804	8804	s8805 Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-


```
In [41]: netflix.groupby(['Genre 2'])['release_year'].count().sort_values(ascending = Fal
```

```
Out[41]: Genre 2
International Movies      1545
Dramas                   744
Independent Movies       572
Comedies                 440
Thrillers                232
Romantic Movies          170
Music & Musicals          119
Sci-Fi & Fantasy          109
Horror Movies            71
Sports Movies            68
Anime Features           49
Faith & Spirituality      49
Cult Movies              44
International TV Shows   41
Children & Family Movies 31
Classic Movies           30
LGBTQ Movies             30
TV Dramas                23
Romantic TV Shows        22
Documentaries            21
TV Comedies              20
Stand-Up Comedy          9
Korean TV Shows          9
Crime TV Shows           8
TV Action & Adventure     7
Docuseries               7
TV Horror                3
Classic & Cult TV         2
Kids' TV                 2
Reality TV               2
TV Sci-Fi & Fantasy        2
Science & Nature TV       1
Spanish-Language TV Shows 1
Stand-Up Comedy & Talk Shows 1
TV Mysteries             1
Name: release_year, dtype: int64
```

```
In [42]: netflix.groupby(['Genre 3'])['release_year'].count().sort_values(ascending = Fal
```

```
Out[42]: Genre 3
International Movies      881
Romantic Movies          431
Thrillers                 278
Music & Musicals          191
Independent Movies       159
Sci-Fi & Fantasy          115
Sports Movies             95
Dramas                    68
LGBTQ Movies              52
TV Dramas                 36
TV Comedies               16
Cult Movies               14
Comedies                  12
Faith & Spirituality      11
Spanish-Language TV Shows 10
TV Mysteries              10
TV Action & Adventure      7
International TV Shows     7
Horror Movies              7
Classic Movies             6
TV Horror                  5
Romantic TV Shows          5
Children & Family Movies   4
Stand-Up Comedy & Talk Shows 3
TV Sci-Fi & Fantasy         3
TV Thrillers               3
Teen TV Shows              3
Korean TV Shows            1
Kids' TV                   1
Name: release_year, dtype: int64
```

```
In [43]: ##no of movies & tv shows released by year arranged in descnding order
netflix.groupby(['release_year'])['release_year'].count().sort_values(ascending=
```

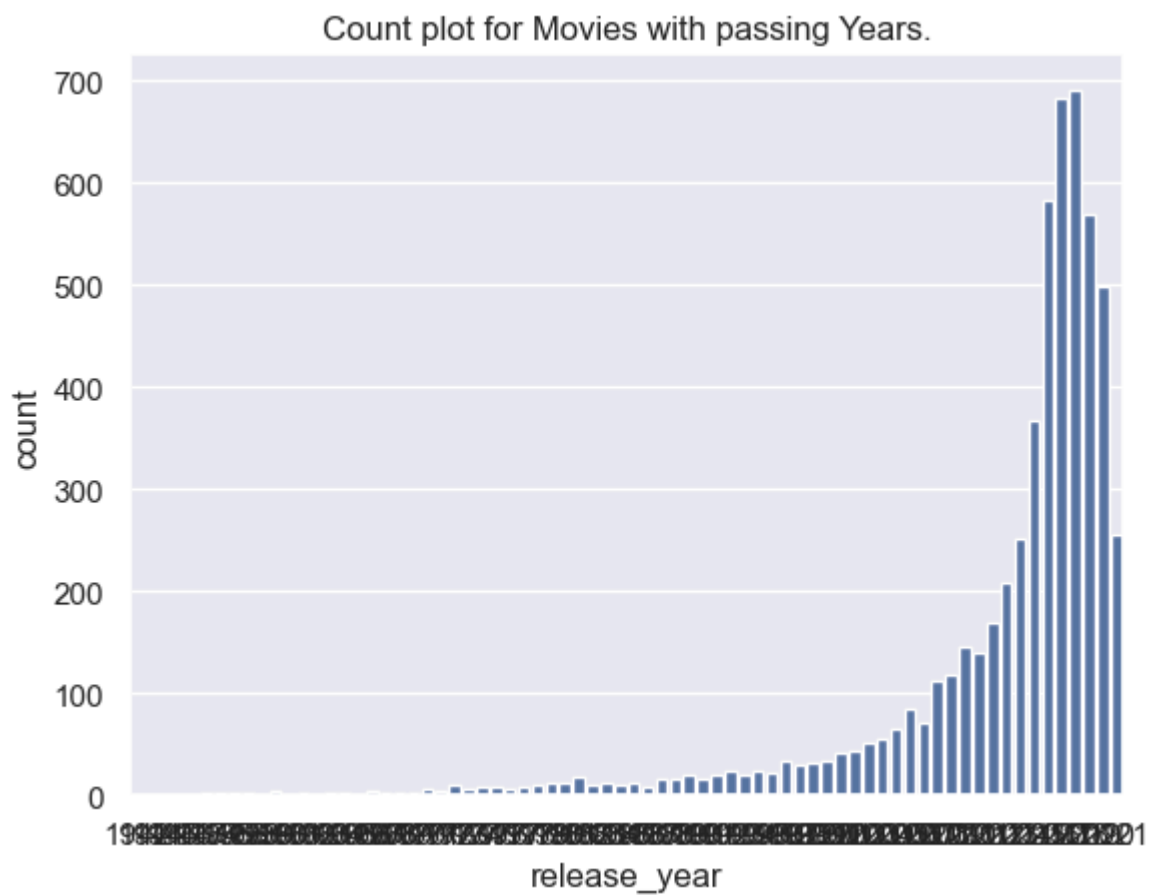
```
Out[43]: release_year
2018      691
2017      684
2016      584
2019      570
2020      499
...
1946        1
1963        1
1961        1
1959        1
1966        1
Name: release_year, Length: 72, dtype: int64
```

Observation:

From the above we can see that with passing years more movies are being made year by year

The above data shows that there is a sudden increase in the creation of movies in year 2016 as compared to year 2015

```
In [44]: sns.countplot(x='release_year', data=netflix).set_title('Count plot for Movies w  
sns.set(rc={'figure.figsize':(60,20)})  
plt.show()
```



```
In [45]: netflix.head()
```

Out[45]:

	Unnamed: 0	show_id	type	title	director	cast	country	date_adde
2	2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	missing	2021-09-2
5	5	s6	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	missing	2021-09-2
6	6	s7	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	missing	2021-09-2
7	7	s8	Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...	United States, Ghana, Burkina Faso, United Kin...	2021-09-2
8	8	s9	TV Show	The Great British Baking Show	Andy Devonshire	Mel Giedroyc, Sue Perkins, Mary Berry, Paul Ho...	United Kingdom	2021-09-2

In [46]:

```
netflix.head()
```


Out[46]:

Unnamed: 0	show_id	type	title	director	cast	country	date_adde
2	2	s3 TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	missing	2021-09-2
5	5	s6 TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	missing	2021-09-2
6	6	s7 Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	missing	2021-09-2
7	7	s8 Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...	United States, Ghana, Burkina Faso, United Kin...	2021-09-2
8	8	s9 TV Show	The Great British Baking Show	Andy Devonshire	Mel Giedroyc, Sue Perkins, Mary Berry, Paul Ho...	United Kingdom	2021-09-2

In [47]:

```
netflix['duration'].unique()
```

```
Out[47]: array(['1 Season', '91 min', '125 min', '9 Seasons', '104 min', '127 min',
'94 min', '161 min', '61 min', '166 min', '147 min', '103 min',
'97 min', '106 min', '111 min', '110 min', '105 min', '96 min',
'124 min', '116 min', '98 min', '115 min', '122 min', '99 min',
'88 min', '100 min', '102 min', '93 min', '95 min', '85 min',
'83 min', '113 min', '13 min', '182 min', '48 min', '145 min',
'87 min', '90 min', '117 min', '128 min', '119 min', '143 min',
'114 min', '118 min', '108 min', '63 min', '121 min', '142 min',
'154 min', '120 min', '82 min', '109 min', '101 min', '86 min',
'229 min', '76 min', '89 min', '156 min', '112 min', '107 min',
'129 min', '92 min', '135 min', '136 min', '165 min', '150 min',
'133 min', '84 min', '140 min', '2 Seasons', '64 min', '59 min',
'139 min', '148 min', '189 min', '141 min', '130 min', '138 min',
'7 Seasons', '132 min', '123 min', '65 min', '68 min', '67 min',
'66 min', '62 min', '69 min', '70 min', '74 min', '131 min',
'46 min', '126 min', '155 min', '4 Seasons', '159 min', '137 min',
'12 min', '77 min', '49 min', '72 min', '78 min', '204 min',
'212 min', '25 min', '47 min', '81 min', '71 min', '149 min',
'15 min', '54 min', '80 min', '224 min', '162 min', '60 min',
'75 min', '79 min', '32 min', '158 min', '164 min', '173 min',
'181 min', '185 min', '73 min', '6 Seasons', '24 min', '51 min',
'151 min', '42 min', '22 min', '134 min', '177 min', '58 min',
'52 min', '53 min', '8 min', '57 min', '28 min', '50 min',
'26 min', '45 min', '171 min', '27 min', '33 min', '44 min',
'29 min', '146 min', '157 min', '203 min', '21 min', '30 min',
'194 min', '55 min', '5 Seasons', '15 Seasons', '233 min',
'237 min', '230 min', '195 min', '253 min', '152 min', '190 min',
'160 min', '23 min', '208 min', '180 min', '144 min', '174 min',
'170 min', '192 min', '209 min', '187 min', '172 min', '186 min',
'11 min', '8 Seasons', '193 min', '176 min', '17 min', '56 min',
'169 min', '40 min', '36 min', '20 min', '168 min', '153 min',
'214 min', '31 min', '163 min', '14 min', nan, '179 min', '38 min',
'3 Seasons', '43 min', '200 min', '196 min', '167 min', '41 min',
'37 min', '35 min', '178 min', '228 min', '18 min', '205 min',
'201 min', '191 min'], dtype=object)
```

```
In [48]: ##netflix_season = netflix[netflix["duration"]=="1 Season"]

##SAHIL
netflix_seasons_data = netflix[netflix['duration'].astype(str).str.contains('Sea
```

```
In [49]: netflix_seasons_data
```

Out[49]:

	Unnamed: 0	show_id	type	title	director	cast	country	date_
2	2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	missing	2021.
5	5	s6	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	missing	2021.
8	8	s9	TV Show	The Great British Baking Show	Andy Devonshire	Mel Giedroyc, Sue Perkins, Mary Berry, Paul Ho...	United Kingdom	2021.
11	11	s12	TV Show	Bangkok Breaking	Kongkiat Komesiri	Sukollawat Kanarot, Sushar Manaying, Pavarit M...	missing	2021.
47	47	s48	TV Show	The Smart Money Woman	Bunmi Ajakaiye	Osas Ighodaro, Ini Dima-Okojie, Kemi Lala Akin...	missing	2021.
...	...	...	...	...	...	...	...	...
8172	8172	s8173	TV Show	Th Eena Meena Deeka Chase Comedy Show	Dheeraj Berry	Sanjay Keni, Priya Raina	India	2018.
8214	8214	s8215	TV Show	The Blue Planet: A Natural History of the Oceans	Alastair Fothergill	David Attenborough	United Kingdom	2015.
8293	8293	s8294	TV Show	The Fear	Michael Samuels	Peter Mullan, Anastasia Hille, Harry Lloyd, Pa...	United Kingdom	2017.
8302	8302	s8303	TV Show	The Five	Mark Tonderai	Tom Cullen, O.T. Fagbenle, Lee Ingleby, Sarah ...	United Kingdom	2017.
8599	8599	s8600	TV Show	Toast of London	Michael Cumming	Matt Berry, Robert Bathurst, Doon	United Kingdom	2017.

Unnamed: 0	show_id	type	title	director	cast	country	date_
------------	---------	------	-------	----------	------	---------	-------

					Mackichan, S...		
--	--	--	--	--	--------------------	--	--

178 rows × 15 columns

```
In [50]: netflix['duration'].head()
```

```
Out[50]: 2      1 Season
5      1 Season
6      91 min
7     125 min
8      9 Seasons
Name: duration, dtype: object
```

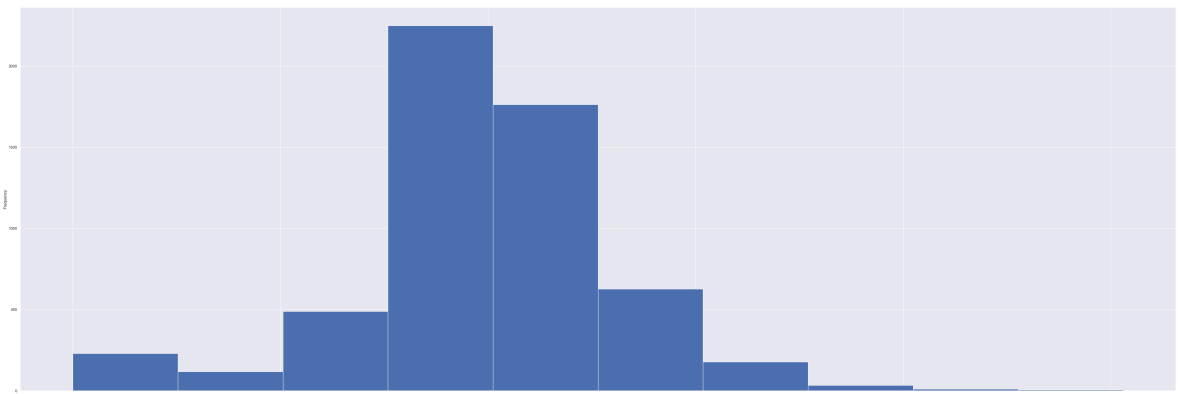
```
In [55]: netflix['duration'] = netflix['duration'].str.replace(' min', '')

netflix['duration'] = pd.to_numeric(netflix['duration'], errors='coerce')

netflix['duration'] = netflix['duration'].fillna(0).astype(int)
```

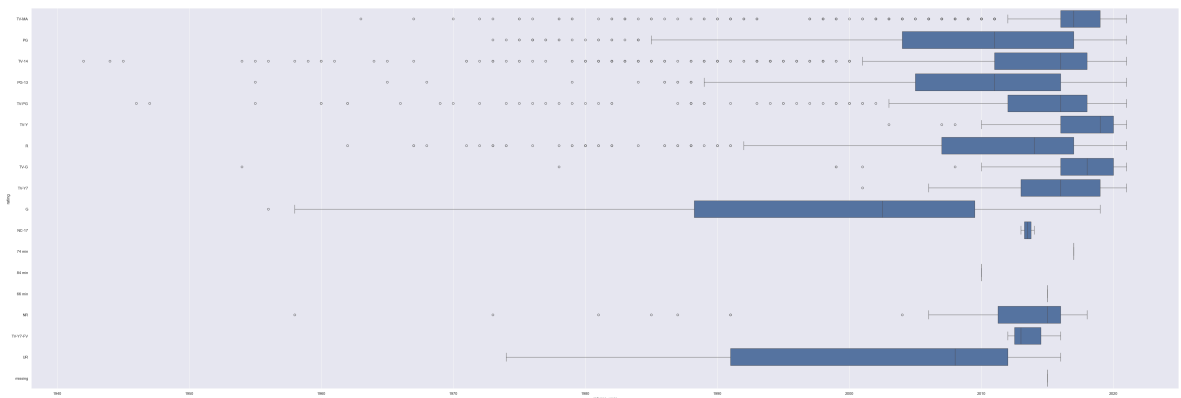
```
In [56]: netflix['duration'].sort_values().plot.hist()
```

```
Out[56]: <Axes: ylabel='Frequency'>
```



```
In [53]: sns.boxplot(x="release_year", y="rating", data=netflix)
```

```
Out[53]: <Axes: xlabel='release_year', ylabel='rating'>
```



```
In [54]: netflix[netflix['Genre 2'] == 'Action'].release_year.groupby(netflix.release_year)
plt.axis('equal')
plt.show()
```

release_year

In []: